

DRY-RUN GUIDE | Matrix.py

Source: Python\3rd_Belt\Matrix\Matrix.py

CORE BEHAVIOR

Creates fixed 2x2 NumPy arrays and assigns their sum, difference, and element-wise product; it does not print those values.

REPRESENTATIVE INPUT

A=[[1,2],[3,4]], B=[[5,6],[7,8]].

STEP TRACE

A+B=[[6,8],[10,12]]; A-B=[[-4,-4],[-4,-4]]; A*B=[[5,12],[21,32]].

EXPECTED OUTPUT / RETURN

No console output: results are assigned only.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\Matrix.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Perform addition, subtraction, and element-wise multiplication on two 2x2 matrices using NumPy
002 | # Add your solution here.
003 |
004 | import numpy as np
005 |
006 | A = np.array([[1, 2], [3, 4]])
007 | B = np.array([[5, 6], [7, 8]])
008 |
009 | add_res = A + B
010 | sub_res = A - B
011 | elem_mult = A * B
012 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.